

Problem-Solution Fit

Project Design Phase | Import Data Using Transform Maps

1. Purpose

Problem-solution fit confirms that the proposed import design addresses the real problems found in the Ideation and Requirement Analysis phases. It links each problem to a design response and checks the fit before development starts.

2. Problem-Solution Fit Canvas

Canvas element	Description
Customer segment	ServiceNow administrators and developers who load external data; data owners who supply files
Jobs to be done	Load files into the right tables accurately, avoid duplicates, and prove what was loaded
Problems / pains	Manual entry, mismatched columns, duplicate records, invalid references, no audit trail
Triggers	New hire list, monthly asset export, legacy system migration, new table go-live
Emotions before	Anxious, frustrated, unsure about accuracy
Emotions after	Confident, in control, able to repeat the task easily
Available solutions	Manual entry, list import, custom scripts
Our solution	Data source + import set + Transform Map with coalesce, scripts and logs
Behaviour change	Admins use one standard method instead of ad-hoc uploads

3. Problem to Design Response

Problem	Design response	Design element
Duplicate records	Match on a unique key and update instead of insert	Coalesce field
Mismatched columns	Explicit mapping between staging and target fields	Field maps
Inconsistent values	Convert and standardise values before saving	Field map scripts
Invalid references	Look up reference records by name or key	Reference lookup
Bad rows in live tables	Validate and skip invalid rows	onBefore script
No audit trail	Log every run and rejected row	Transform history, import log
Repeated effort	Reuse data source and map	Saved Transform Map

4. Fit Validation

Check	Result
Does the design solve the stated problem?	Yes: every pain point has a matching design element

Check	Result
Is it feasible on a standard instance?	Yes: uses built-in import features only
Will users adopt it?	Likely: one method, reusable, with visible results
Is anything missing?	Web service imports deferred to a later phase

5. Design Principles

- Separate raw data (staging) from live data (target).
- Never write to a live table without a mapping and a unique key.
- Keep transformation logic simple, commented and testable.
- Log everything so results can be verified.
- Design for reuse across future files.